



NATIONAL UNIVERSITY OF MODERN LANGUAGES

Computer Networks Lab No 7

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PROGRAM: BSSE (5th Semester)

COURSE: Computing Networks

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LAB 7: Cisco Discovery Protocol (CDP) and Servers

Task 1: CDP Neighbor Information

In this task, CDP is used to discover directly connected Cisco devices and their details.

Commands Used

Router# show cdp neighbors

Router# show cdp neighbors detail

These commands display:

- Neighbor device IDs
- Interface details
- Platform and capability information
- IP addresses of connected devices

```
Router>ena
Router>enable
Router#config
Router#configure t
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#inter
Router(config)#interface g
Router(config)#interface gigabitEthernet 0/0
Router(config-if)#ip add
Router(config-if)#ip address 192.168.10.1 255.255.255.0
Router(config-if)#no sh
Router(config-if)#no shutdown
```

```
Router#
Router#show cdp neighbors
Capability Codes: R - Router, T - Trans Bridge, S - Source Route Bridge
                  S - Switch, H - Host, I - ICMP, r - Repeater, P - Phone
Device ID    Local Interface    Holdtime    Capability    Platform    Port ID
Switch      Gig 0/0          178         2             2960        Gig 0/1
Router#show cdp neighbors detail

Device ID: Switch
Entry address(es):
Platform: cisco 2960, Capabilities: Switch
Interface: GigabitEthernet0/0, Port ID (outgoing port): GigabitEthernet0/1
Holdtime: 163

Version :
Cisco IOS Software, C2960 Software (C2960-LANBASEK9-M), Version 15.0(2)SE4, RELEASE SOFTWARE
(fx)
Technical Support: http://www.cisco.com/techsupport
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Compiled Wed 26-Jun-13 02:49 by nguyen

advertisement version: 2
Duplex: full
Router#
```

Task 2: Disable CDP Globally

To disable Cisco Discovery Protocol on the entire router, the following command is used:

Router# no cdp run

This command stops CDP advertisements and prevents the router from sharing device information.

```
Router>en
Router>enable
Router#no cdp run
      ^
% Invalid input detected at '^' marker.
Router#
```

Task 3: Disable CDP on a Specific Interface

CDP can also be disabled on a single interface instead of the whole router.

Configuration Commands

Router(config)# interface FastEthernet 0/1

Router(config-if)# no cdp enable

This ensures CDP is not active on the selected interface only.

```
Router#enable
Router#conf
Router#configure t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#inter
Router(config)#interface g
Router(config)#interface gigabitEthernet 0/1
Router(config-if)#no cdp en
Router(config-if)#no cdp enable
Router(config-if)#exit
Router(config)#exit
```

Switch configuration

```
Switch>enable
Switch>configure t
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)# inter
Switch(config)# interface vlan 1
Switch(config-if)#ip add
Switch(config-if)#ip address 192.168.10.2 255.255.255.0
Switch(config-if)#no sh
Switch(config-if)#no shutdown

Switch(config-if)#
%LINK-5-CHANGED: Interface Vlan1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan1, changed state to up

Switch(config-if)#exit
Switch(config)#ip default-gateway 192.168.10.1
Switch(config)#cdp run
Switch(config)#host
Switch(config)#hostname S1
S1(config)#end
S1#
%SYS-5-CONFIG_I: Configured from console by console

S1#write memory
Building configuration...
[OK]
S1#
S1#
```

CDP Checking

```
SI>sh
SI>show cdp
Global CDP Information:
  Sending CDP packets every 60 seconds
  Sending a holdtime value of 180 seconds
  Sending CDPv2 advertisements is enabled

SI#show cdp neighbors
Capability Codes: R - Router, T - Trans Bridge, B - Source Route Bridge
                  S - Switch, H - Host, I - IGMP, r - Repeater, F - Phone
Device ID      Local Interface  Holdtime  Capability  Platform  Port ID
Router        Gig 0/1         171      R           C2900     Gig 0/0
SI#show cdp neighbors detail

Device ID: Router
Entry address(es):
  IP address : 192.168.10.1
Platform: cisco C2900, Capabilities: Router
Interface: GigabitEthernet0/1, Port ID (outgoing port): GigabitEthernet0/0
Holdtime: 152

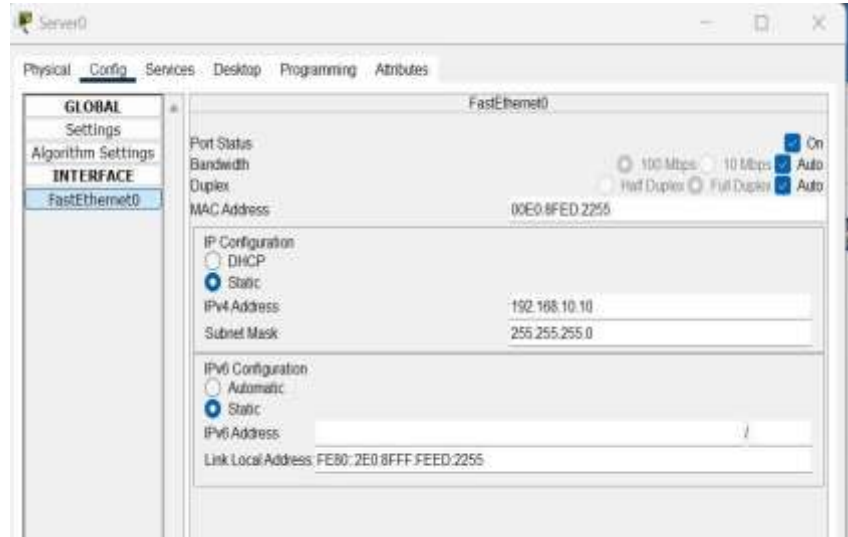
Version :
Cisco IOS Software, C2900 Software (C2900-UNIVERSALK9-M), Version 15.1(4)M4, RELEASE SOFTWARE
(fx)
Technical Support: http://www.cisco.com/techsupport
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advertisement version: 2
Duplex: full
SI#
```

Server configuration

Step 1 — Give the Server a Static IP

1. On the **Server** → **Config > FastEthernet**



Step 2 — Enable DHCP Service

1. **Server > Services > DHCP**



Step 3 — Set PCs to Obtain IP via DHCP

For **PC1** and **PC2**:

Desktop > IP Configuration > select **DHCP**

